Ex:No: 7 Date: 15/4/26	**Implementation of Linear Regression Using mtcars Dataset**

Problem Description:

The mtcars dataset contains specifications of different car models, including fuel efficiency and engine characteristics. To understand how engine and vehicle attributes influence fuel efficiency, linear regression can be used to model the relationship between **miles per gallon (mpg)** and predictors such as **engine displacement (disp)**, **horsepower (hp)**, and **weight (wt)**.

Objective:

To implement a linear regression model in R to analyze the relationship between fuel efficiency and vehicle attributes, interpret the regression coefficients and model summary, and visualize the regression results for better understanding of the model.

Procedure:

- **Step 1: Load Required Libraries**

  We load the necessary libraries for implementing linear regression.

```
# Load required libraries
library (caret)
library (ggplot2)
```

- **Step 2: Load and Prepare the Dataset**

  We will use the built-in mtcars dataset and explore it.

```
# Load the dataset
data (mtcars)

# Display first few rows
head (mtcars)
```

Dataset Explanation:

mpg (Miles per gallon) → Dependent Variable (Y)

hp (Horsepower) → Independent Variable (X)

We will predict mpg based on hp using linear regression.

>head (mtcars)

	mpg	cyl	disp	hp	drat	wt	qsec
Mazda Rx4	21.0	6	160	110	3.90	2.620	16.46
mazda Rx4 wag	21.0	6	160	110	3.90	2.875	17.02
Datsun 710	22.8	4	108	93	3.85	2.320	18.61
Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44
Hornet sportabout	18.7	8	360	175	3.15	3.440	17.02
valiant	18.1	6	225	105	2.76	3.460	20.22

vs	am	gear	carb
0	1	4	4
0	1	4	4
1	1	4	1
1	0	3	1
0	0	3	2
1	0	3	1

- **Step 3: Split the Dataset into Training and Testing Sets**
  We split the dataset into 70% training and 30% testing.

  # Split data into training (70%) and testing (30%)

```
train_index <- createDataPartition(mtcars$mpg, p=0.7,
                                   list=FALSE)
train_data <- mtcars[train_index,]
test_data <- mtcars[-train_index,]
```

- **Step 4: Simple Linear Regression (mpg vs hp)**
  We build the linear regression model using lm() function.

  # Train linear regression model

```
lm_model <- lm(mpg ~ hp, data = train_data)
```

  # Print model summary

```
summary(lm_model)
```

- **Step 5: Make Predictions on Test Data**
  We use the trained model to make predictions on the test dataset.

  # Make predictions on test data

```
predictions <- predict(lm_model, test_data)
```

  # Display predicted values

```
print(predictions)
```

```
> summary (model)
Call:
lm (formula = mpg ~ hp, data = train_data)
Residuals:
   Min      1Q     Median     3Q      Max
-5.7121  -2.1123  -0.2212  1.5876  8.9234
Coefficients:
            Estimate  Std. Error  t value  Pr(>|t|)
(intercept) 30.09886  1.63392     18.42    <2e-16 ***
hp          -0.06823  0.01196     -5.70    4.9e-06 ***
Signif. Codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ',
Residual standard error: 3.863 on 30 degrees of freedom
Multiple R-squared : 0.6204
Adjusted R-squared : 0.6031
F-statistic : 35.95 on 1 and 30 DF, p-value : 4.919e-06.


> print (predictions)
mazda RX4 wag Duster 360 chrysler Imperical dodge challe
22.24053  12.60026  13.6714.0  19.3841
lotus Europe  Ford Pantera L  Ferrari Dino  Porsche 914-2
22.02630  11.24348  17.59892  23.59731
```

- **Step 6: Evaluate Model Performance**

  We compute Mean Squared Error (MSE) and R-squared for evaluation.

  # Compute Mean Squared Error (MSE)

```
mse <- mean ((test_data$mpg - predictions)^2)
cat ("Mean Squared Error: ", mse, "\n")
```

  # Compute R-squared value

```
R-squared <- cor(test_data$mpg, predictions)^2
cat ("R-squared: ", R-squared, "\n")
```

- **Step 7: Visualize Classification Results**

  We plot the regression line on a scatter plot.

  # Scatter plot with regression line – hp vs mpg

```
ggplot (train_data, aes (x= hp, y=mpg)) +
geom_point (color = "blue", size=3) +
geom_smooth (method = "lm", col="red") +
ggtitle ("Linear Regression: Horsepower vs MPG")
```

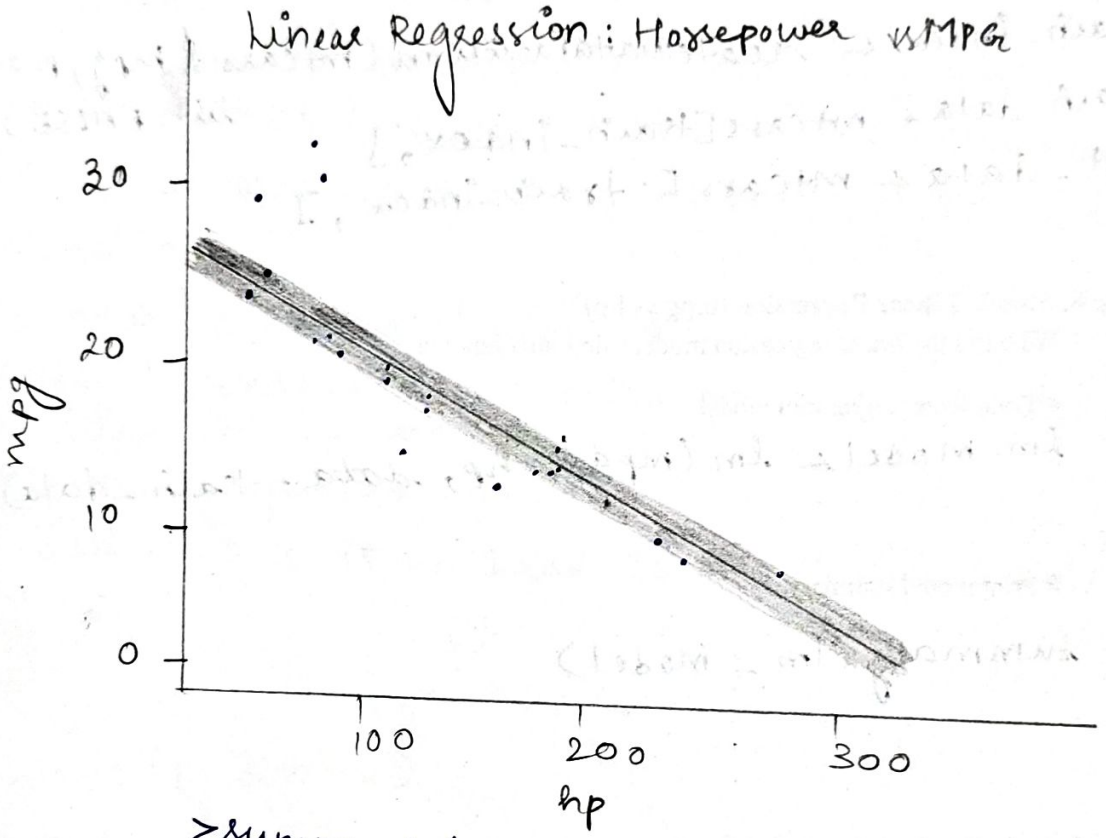
- **Step 8: Multiple Linear Regression - mpg – wt, hp, disp**

```
model_multiple <- lm (mpg ~ wt + hp + disp, data=mtcars)
```

  # Model summary

```
summary (model_multiple)
```

```
>cat ("Mean squared Error:", mse, "\n")
Mean Squared Error: 15.20511
>cat ("R-squared: ", r-squared, "\n")
R-squared: 0.6163636
```




```
>summary (model_multiple)
Call:
lm (formula = mpg ~ wt + hp + disp, data = mtcars)
Residuals:
    Min      1Q   Median      3Q     Max
-3.5405 -1.6701 -0.4264  1.1320  5.4996
Residual Standard error: 2.639 on 28 degrees of freedom
Multiple R-squared: 0.8268
Adjusted R-squared: 0.8083
F-statistic: 44.51 on 3 and 28 DF, P-value: 1.19e-10
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 37.105505  2.110815  17.579  <2e-16 ***
wt          -3.800891  1.066191  -3.565  0.00133 **
hp          -0.031157  0.011436  -2.724  0.01097 *
disp        -0.000937  0.010350  -0.091  0.92851
```

- **Step 9: Multiple Linear Regression - Prediction**

```
# Predicted mpg values
predicted_mpg <- predict(model_multiple)
head(predicted_mpg)
```

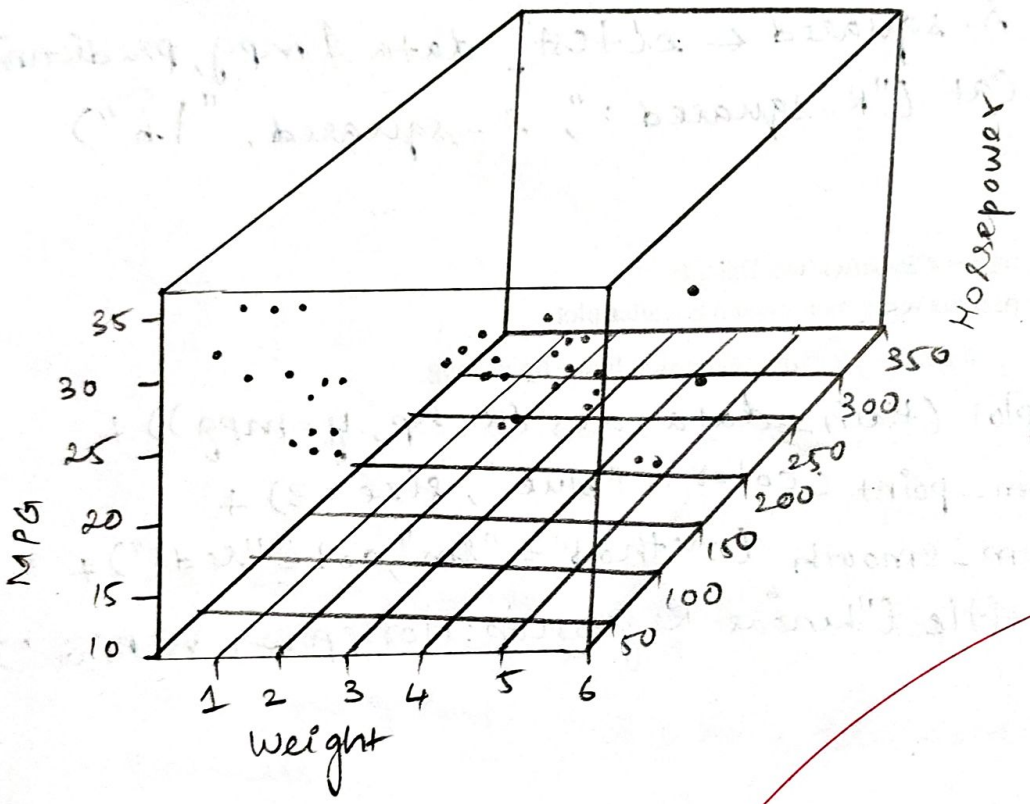
- **Step 10: 3D Visualization (mpg vs wt and hp)**

```
library(scatterplot3d)
scatterplot3d(mtcars$wt, mtcars$hp, mtcars$mpg,
  xlab = "weight",
  ylab = "Horsepower",
  zlab = "MPG",
  main = "3D plot of MPG vs Weight and Horsepower")
```

Result:

Thus, the regression analysis demonstrates that vehicle weight, horsepower, and engine displacement have a significant negative effect on fuel efficiency. The multiple linear regression model provides better explanatory power than simple regression and successfully predicts miles per gallon based on vehicle characteristics.

> head (predicted_mpg)

mazda Rx4	Mazda Rx4 wag	Datsun 710
23.57003	22.60080	25.28868
Hornet 4 Drive	Hornet Sportabout	Valiant
21.21667	18.24072	20.47216

3D plot of MPG vs weight and Horsepower